User Manual

Digital Buffer

DB-50-F-A
DB-50-R-A

User Manual – Version v1.1.0
18 March 2026

Contents

1 General Information

This chapter provides essential information on the purpose of this manual, the manufacturer, regulatory compliance, warranty conditions, and the conventions used throughout this document.

1.1 About This Manual

This manual provides instructions for the installation, configuration, operation, maintenance, and disposal of the Digital Buffer. It forms an integral part of the product and must remain accessible to the personnel responsible for the device. If the device is transferred to a third party, all accompanying documentation must be delivered with it.

This manual contains proprietary information and may not be reproduced or distributed, in whole or in part, without prior written consent of the manufacturer.

> Before performing any operation, the entire manual must be read carefully and the relevant sections reviewed in detail.

1.1.1 User Qualification

The device is intended for use by qualified personnel familiar with laboratory electronic instrumentation and responsible for the installation, operation, and maintenance of the device throughout its service life. It is not a toy and must not be used by children under 14 years of age. Persons with reduced physical, sensory, or cognitive abilities may operate the device only under supervision by qualified personnel.

1.1.2 Revision History

Document Revision: v1.1.0- 18 March 2026

A certified copy of this manual is stored in the manufacturer's technical file for the legally required period (10 years). During this period, documentation may be requested from the manufacturer.

1.2 Symbols and Conventions

The following graphical symbols and textual conventions may appear in this manual or on the device.

> Provides important supplementary information.

> Indicates procedures that, if not observed, may result in damage to the device or associated equipment.

> Indicates procedures that, if not observed, may result in injury or serious damage.

(a) General hazard symbol **(b)** Mandatory reading of the manual

Figure 1.1: The device may include the pictograms shown above.

All safety symbols must remain clearly visible throughout the lifetime of the device.

1.3 Manufacturer Information

Manufacturer:	FLIM LABS S.r.l.
Address:	Via della Farnesina, 3 00135 Rome (RM), Italy
Telephone:	+39-329 761540
E-mail:	info@flimlabs.com
Website:	www.flimlabs.com

1.4 Declaration of Conformity

The device is supplied with a Declaration of Conformity drawn up in accordance with applicable European legislation. Before use, verify that the Declaration of Conformity is present. A copy of the Declaration of Conformity is shown in Document 1.1.

1.5 Product Label

The product label affixed to the device contains the required regulatory markings and identification information including:

- Product name and model
- Manufacturer identification
- Regulatory markings
- Electrical specifications

An example product label is shown in Figure 1.2. Disposal requirements associated with the WEEE marking are described in Section 7.2.

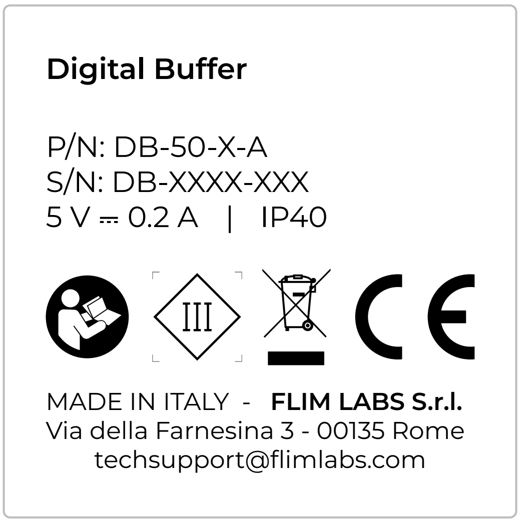

Figure 1.2: Product label

Safety pictograms performing a protective function must not be removed, covered, or damaged. If the label becomes illegible or damaged, contact authorized service for replacement.

1.6 Warranty and Liability

1.6.1 Warranty

The manufacturer provides 24 months warranty for private customers and 12 months warranty for commercial customers. The warranty period begins on the date of delivery.

EU Declaration of Conformity

Device Information	
Type:	Digital Buffer
Model:	Digital Buffer
Part Number:	DB-50-F-A DB-50-R-A
Serial Number:	As indicated on the product label

Manufacturer Information	
Manufacturer:	FLIM LABS S.r.l.
Address:	Via della Farnesina, 3 00135 ROME (RM), Italy
Telephone:	+39 – 329 761540
E-mail:	info@flimlabs.com
Website:	www.flimlabs.com

The present declaration of conformity is issued under the sole responsibility of the manufacturer. The device complies with the following directives and regulations:

Applicable Directives and Regulations

- Directive 2014/35/EU, known as the "Low Voltage Directive"
- Directive 2014/30/EU, known as the "Electromagnetic Compatibility Directive"
- Directive 2011/65/EU, known as "RoHS"
- Delegated Directives (EU) 2015/863 and (EU) 2017/2102 amending Directive 2011/65/EU
- Directive 2012/19/EU, known as "WEEE"
- Regulation (EU) 2023/988 on General Product Safety

The manufacturer undertakes to provide, upon a duly reasoned request from the national authorities, all relevant information regarding the device.

Signed for and on behalf of FLIM LABS S.r.l.:

Place and Date	**Signature**
Rome, 24/10/2025	Alessandro Rossetta, CEO

Document 1.1: EU Declaration of Conformity

The warranty covers defects in materials and workmanship under normal and intended use. It is valid only if the device is operated and maintained according to this manual. Unauthorized modification, improper use, failure to follow instructions, or repair by non-authorized personnel voids the warranty. Full warranty terms are available in the manufacturer's General Terms and Conditions published on the official website.

To submit a warranty claim, provide the following:

- Product type/version
- Proof of purchase (including date)
- Detailed description of the issue

1.6.2 Limitation of Liability

The device has been designed, manufactured, and tested in accordance with applicable European directives and internal quality standards. The manufacturer is liable exclusively for defects attributable to manufacturing faults or non-conformity with declared specifications, within the limits defined in the warranty terms.

The manufacturer shall not be liable for damages resulting from improper use, including but not limited to:

- Use outside the intended purpose
- Failure to follow this manual or applicable safety instructions
- Installation or operation by unqualified personnel
- Operation outside specified electrical or environmental limits
- Use of incompatible or non-compliant input signals or connected equipment
- Unauthorized modification, repair, or use of non-original spare parts
- Integration into systems not compliant with applicable safety regulations

The user is responsible for ensuring that the device is installed, operated, and integrated in accordance with this manual and all applicable local, national, and international regulations.

Except where otherwise required by mandatory law, the manufacturer shall not be liable for indirect, incidental, or consequential damages, including but not limited to loss of data, loss of profit, or system downtime.

2 Device Description

This section describes the functional characteristics, intended application, and technical features of the Digital Buffer.

2.1 Functional Description

The Digital Buffer is a compact electronic module designed for the conversion and distribution of pulse signals into standardized digital signals. It is primarily used in scientific instrumentation, timing, event counting, and fluorescence lifetime imaging microscopy (FLIM) applications, where input digital signals must be adapted for reliable downstream processing.

The product is housed in a compact aluminum enclosure suitable for benchtop laboratory use or integration into portable measurement systems. The metallic enclosure provides mechanical robustness and adequate electromagnetic shielding during operation within the specified limits. The device is equipped with one SMA input and two digital outputs implemented via SMA coaxial connectors. Power is supplied via USB Type-C at 5 V DC.

The Digital Buffer is designed to receive a pulsed digital input signal via SMA, with input voltage levels typically between 1.5 V and 5 V on an internal 50 Ω load. The input signal is internally converted and replicated on two standard LVTTL digital outputs at 2.5 V on 50 Ω, enabling duplication and distribution of the signal to data acquisition devices, timing modules, or other electronic instrumentation. The operating principle is based on active logic-level conversion and distribution of the signal to two electrically independent output channels while maintaining the timing characteristics of the original pulse. The device is intended for integration into existing measurement chains and does not require configuration.

The product is available in two functional variants, which differ only in the input edge detection mode:

- **DB-50-F-A**: optimized for falling-edge input signals
- **DB-50-R-A**: optimized for rising-edge input signals

The variants do not differ in electronic architecture, operating principle, or safety requirements. All modules feature:

- Compact digital buffer module
- One SMA pulse input
- Two SMA digital outputs
- Standard LVTTL 2.5 V outputs on 50 Ω
- USB Type-C 5 V power supply
- Compact aluminum enclosure suitable for system integration
- No user configuration required
- Suitable for signal duplication and distribution
- Power operation compliant with SELV requirements

2.2 Intended Use

The Digital Buffer is intended as a compact electronic module for the conversion and distribution of digital signals by professional users in private research laboratories, companies, and universities. It may be used as a stand-alone unit or integrated into custom measurement systems and instrumentation.

2.3 Reasonably Foreseeable Misuse

The following uses are not permitted:

- Use outside described applications
- Use with incompatible components

- Operation outside specified limits
- Unauthorized modification
- Use with incorrect power supply

2.4 Components Overview

Figure 2.1 shows the key components of the device, including:

1. USB Type-C power input
2. SMA signal input
3. SMA output 1 and 2
4. Mounting holes for mechanical fixing

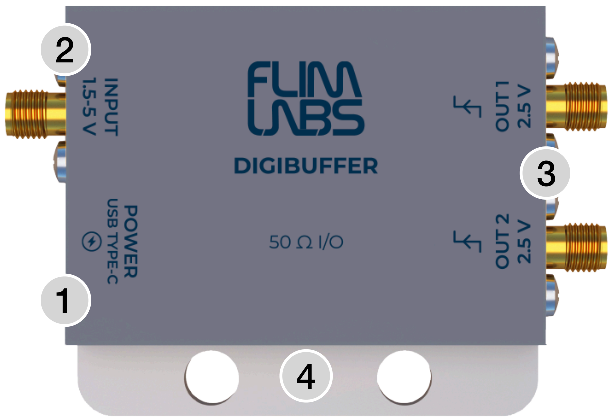

Figure 2.1: Front and back view showing the key components of the Digital Buffer.

2.5 Mechanical Drawings and Dimensions

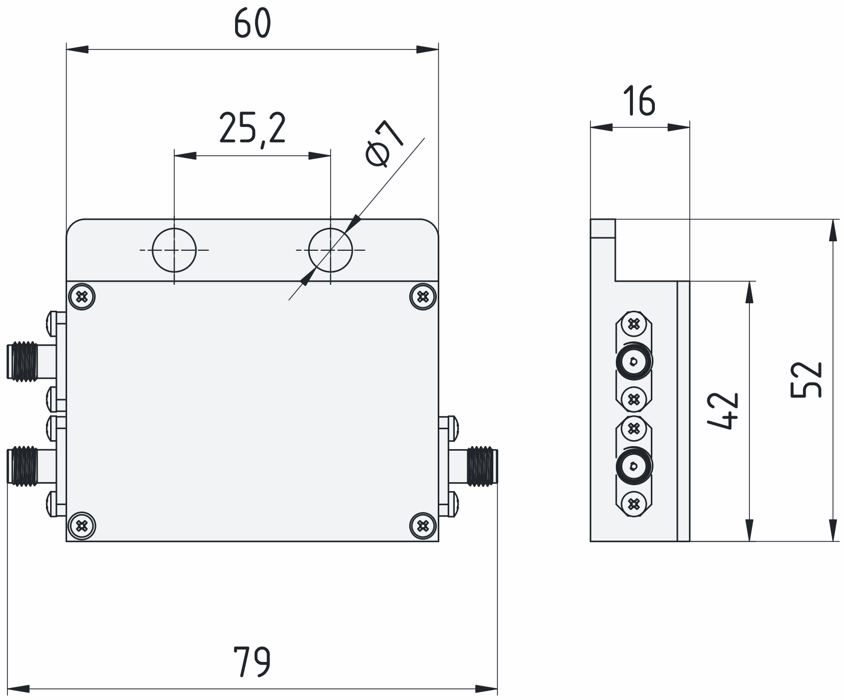

Figure 2.2: Technical drawing of the Digital Buffer.

2.6 Technical Specifications

The electrical, signal, and environmental limits referenced throughout this manual are summarized in this section.

Item	DB-50-F-A	DB-50-R-A
Electrical Specifications		
Electrical safety class	Class III	
Power supply	USB Type-C, 5 V DC	
Current consumption (max)	0.2 A	
Power consumption (max)	1 W	
Signal Specifications		
Input connections	1x SMA	
Input polarity	negative	positive
Minimum input pulse width	>4 ns	
Minimum input pulse height (abs)	1.5 V	
maximum input pulse height (abs)	5 V	
Output connections	2x SMA	
Output signal	SMA: 2.5 V digital LVTTL at 50 Ω	
Output pulse width	>4 ns	
Maximum repetition frequency	100 MHz	
Impedance	50 Ω	
Physical Specifications		
Dimensions (L x W x H)	60 mm x 42 mm x 14 mm	
Weight	80 g	
Protection rating	IP40	
Environmental Specifications		
Operating temperature	0 °C to +40°C	
Storage temperature	-20 °C to +70°C	
Humidity	up to 80% non-condensing	

Table 2.1: Technical specifications of the Digital Buffer.

3 Safety Information

3.1 General Safety Instructions

Before installation and operation, ensure that this manual is complete and legible. If any part of the documentation is missing or unclear, contact the manufacturer before proceeding.

The device must be used only as described in this manual. Any operation not explicitly described in this manual is considered improper use.

This manual complements applicable national and local safety regulations. If local regulations impose stricter requirements, those regulations take precedence.

Despite the implemented safety measures, residual risks may remain if the device is not used as specified.

3.2 Electrical Safety

- Operate the device only with the specified power supply.
- Use undamaged, properly rated cables.
- Disconnect power before modifying signal connections.
- Do not open the enclosure. There are no user-serviceable internal components.
- Keep away from water sources and other liquids.
- Unauthorized repair or modification is not permitted.

3.3 Environmental and Operating Conditions

The device is designed for indoor laboratory use only. Operate the device only within the specified environmental limits:

- Operating temperature: 0°C to +40°C
- Storage temperature: −20°C to +70°C
- Relative humidity: up to 80% non-condensing
- Stable, vibration-free surface

Do not operate the device in explosive atmospheres or in the presence of flammable gases.

Ensure adequate ambient lighting during installation and adjustment to avoid connection errors.

3.4 Modification and Improper Use

The device must not be modified, altered, or repaired by unauthorized personnel. Use outside the intended application, operation beyond specified electrical limits, or failure to follow the procedures described in this manual may result in malfunction and void the warranty.

3.5 Documentation Retention

This manual must be kept accessible to personnel responsible for installation, operation, and maintenance. A replacement copy may be requested from the manufacturer if required.

4 Setup

This section describes the correct procedure for unpacking, installing, and configuring the device before operation.

4.1 Scope of Delivery

- Digital Buffer
- USB Type-C power cable
- Quick start guide
- Declaration of conformity

Verify that all items are present before proceeding with installation.

4.2 Unpacking and Inspection

The device is supplied in protective packaging to protect it during storage and transport. Handle the device carefully during unpacking to avoid mechanical damage.
After removing the packaging:

- Inspect the device and accessories for visible damage.
- Check for dents, deformation, cracks, or signs of impact.
- Verify that connectors and cables are intact.

If any damage, missing parts, or abnormalities are detected, stop the installation and contact the manufacturer before proceeding.
Dispose of packaging materials responsibly and keep them out of reach of children.
The device is compact and lightweight and may be moved manually. When handling, support it securely to prevent accidental dropping.

4.3 Installation Requirements

Before installation and operation, ensure that the environmental conditions specified in Section 3.3 are met. Children must not have access to the installation or storage area. The device requires a regulated 5 V DC power source via USB Type-C.

> Before connecting the device, verify that the input signal characteristics are compatible with the technical specifications of the CFD (see Table 2.1). Improper signal conditions may result in malfunction or unreliable operation. An overview of the required input signal checks is provided in Figure 4.1.

4.4 Mechanical Installation

Place the device on a stable laboratory surface with adequate space for cable routing and ventilation.
Ensure that:

- The device is positioned to prevent mechanical stress on connectors.
- Cables are routed safely and do not create tripping hazards.
- The power supply location is dry and away from water sources (e.g., sinks or splash zones).

Remove any protective dust or moisture covers before operation.

4.5 Electrical Connections

All electrical connections must be performed by qualified personnel who have read this manual. Connector locations are shown in Figure 2.1.
Proceed as follows:

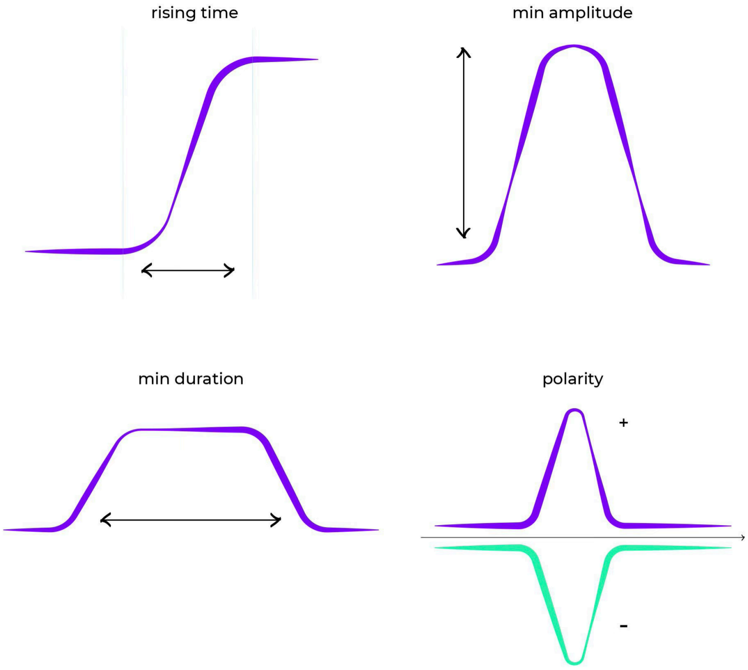

Figure 4.1: Verify that the input signal characteristics.

1. **Power Connection**:
    a) Connect the USB Type-C cable to the device power input.
    b) Connect the other end to a suitable 5 V DC power source such as a computer USB port, a power bank, or a certified 5 V DC adapter.
    c) Verify visually that the LED power indicator confirms correct device power-up.
2. **Signal Connection**:
    a) Connect the digital input signal to the SMA input connector.
    b) Connect one or both SMA outputs to the downstream measurement or processing devices.

4.6 Initial Configuration

After completing all electrical connections, no further configuration is required.

5 Operation

This section describes the functional behavior of the Digital Bufferduring use, including its operating principle, normal operating conditions, adjustable parameters, and troubleshooting guidance.

5.1 Principle of Operation

The Digital Buffer receives a pulse digital input signal and provides two duplicated digital output signals in standardized LVTTL format. The device converts the incoming digital signal level and actively distributes it to two electrically independent SMA output channels while preserving the timing characteristics of the original pulse. The device provides:

- One digital input (SMA)
- Two digital outputs (SMA)
- USB Type-C power input
- LVTTL output signals at 2.5 V on 50 Ω

Typical applications include duplication and conditioning of pulse signals in scientific instrumentation, timing systems, event counting chains, and FLIM-related measurement setups.

5.2 Normal Operation

Do not operate the device if it is damaged or shows signs of unsafe condition.

Once the power supply, input signal, and output connections have been correctly established and the initial configuration has been completed, the Digibuffer operates autonomously without further user intervention. Correct functionality depends on proper system integration, and the user is responsible for ensuring that the complete measurement setup meets applicable safety and performance requirements.

5.3 Parameter Adjustment

The device does not require user-adjustable parameters for normal operation.

After installation, verify stable and consistent output behavior under expected operating conditions. Typical integration examples are shown in Figures 5.1.

5.4 Troubleshooting

Common faults and corrective actions are summarized in Table 5.1.

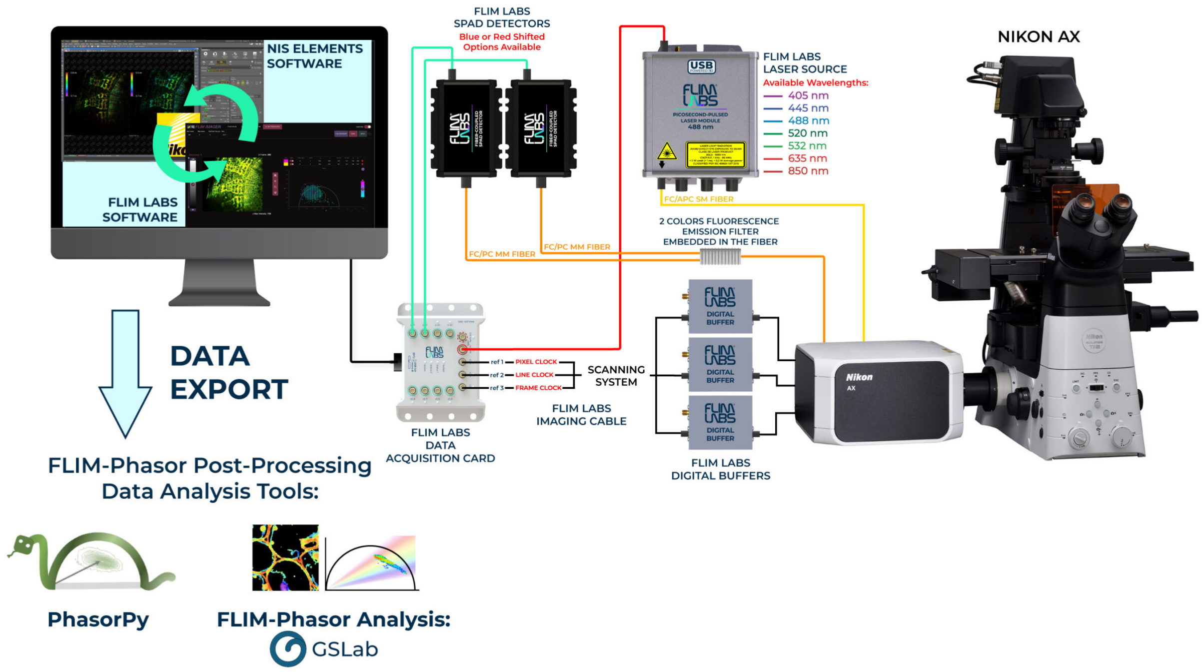

Figure 5.1: Exmaple of employing the Digital Buffer for signal conditioning in a microscope setup.

Table 5.1: Overview of common problems and recommended corrective actions.

Problem	Possible Cause	Recommended Action
Power and Device Status		
No output signal	The device is not connected to a power source	Connect the device to a suitable power source according to the instructions.
The device does not power on via USB	Insufficient power supplied by the USB port	Use a USB 3.0 port or another USB power source capable of supplying 900 mA at 5 V.
The device does not power on	Defective USB cable	Replace the USB Type-C cable and verify proper connector seating.
Input Signal		
No output signal	The input signal does not meet the requirements specified in the technical data	Check the input signal with an oscilloscope or another suitable instrument and verify compliance with the input specifications.
No output signal	Input signal amplitude too low or signal type not suitable	If available, use a suitable digital signal or use a signal conversion or conditioning device.
Output signal unstable	Noise on the input signal or improper cabling	Improve signal quality, reduce cable length, and ensure proper shielding and grounding.

Problem	Possible Cause	Recommended Action
Output Signal		
No signal detected by external devices	Incorrect output connection	Verify that both SMA outputs are correctly connected to the intended external devices.
Output signal distorted	Improper impedance matching or excessive cable length	Use short 50 Ω coaxial cables and ensure proper termination of the signal path.

If any unexpected or potentially hazardous anomaly occurs during startup or operation, switch off the device immediately and disconnect it from the power source. Contact technical support before further use.

6 Maintenance and Service

This section describes the cleaning, preventive maintenance, corrective maintenance, and authorized service procedures for the device. All maintenance activities must be performed by qualified personnel who have read and understood this manual.

> Before performing any cleaning or physical maintenance operation, the device must be switched off and disconnected from the power source.

Maintenance must be carried out in compliance with this manual and applicable workplace safety regulations. The maintenance area should provide adequate illumination; a minimum lighting level of 200 lux is recommended.

6.1 Cleaning

The device requires periodic external cleaning to ensure reliable operation.

- Remove dust or dirt from the exterior surfaces using a dry or slightly damp lint-free cloth.
- A mild, neutral detergent may be used if necessary.
- Do not allow liquids to enter connectors or internal components.
- Avoid the use of solvents, aggressive chemicals, or abrasive materials.

For connectors and hard-to-reach areas (e.g., USB ports), a soft-bristled brush or compressed air may be used.

6.2 Preventive Maintenance

The device does not require routine internal maintenance.
Periodically inspect:

- External surfaces for mechanical damage
- Connectors and cables for wear or deformation
- Product label for legibility. See Section 1.5 for label content and handling requirements.

If any abnormal condition is detected, discontinue use and contact technical support.

6.3 Service and Repairs

The device contains no user-serviceable internal components. In the event of malfunction, accidental damage, or improper operation:

- Switch off and disconnect the device.
- Do not attempt internal repair.
- Contact the manufacturer's technical service for instructions.

Repairs or modifications must be carried out exclusively by FLIM LABS S.r.l. or authorized personnel. Use only original FLIM LABS spare parts. Further warranty conditions are described in Section 1.6.1.

> Unauthorized intervention or the use of non-original components may compromise safety, performance, and warranty coverage.

7 Decommissioning and Disposal

This section describes procedures for removing the device from service and disposing of it in accordance with applicable regulations.

7.1 Decommissioning Procedure

The device is designed for long-term laboratory use. When it reaches the end of its technical or operational life, it must be decommissioned to prevent further use.

Decommissioning requires:

- Disconnecting the device from all power sources.
- Removing all signal connections.
- Ensuring that the device cannot be unintentionally reused.

The same procedure applies in cases of:

- Permanent withdrawal from service
- Storage without intended future use
- Final disposal

Reuse of components after final decommissioning is not permitted unless explicitly authorized by the manufacturer.

7.2 Disposal

The device must be disposed of in accordance with applicable local and European regulations governing electrical and electronic equipment. In accordance with Directive 2012/19/EU (WEEE), the device must not be disposed of with household waste. The product label includes the WEEE symbol indicating the requirement for separate collection of electrical and electronic equipment (see Section 1.5). Contact your local waste management authority or certified collection center for proper disposal procedures.

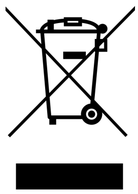

Figure 7.1: WEEE symbol